



Kinesis Data Streams vs. Amazon MSK

Choosing the Right Streaming Service

AWS Machine Learning Associate • Part 5

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What We'll Cover



What they have in common



Amazon Kinesis Data Streams



Amazon MSK



Key differences



When to use Kinesis Data Streams



When to use Amazon MSK

What They Have in Common



- Both are fully managed, real-time streaming services on AWS
- Both durably store streaming data for a retention window, so consumers can process it independently
- Both can feed the same downstream services: Lambda, S3, Redshift, and ML pipelines

The real decision comes down to the technology underneath, and how much control you want.

Amazon Kinesis Data Streams



AWS-Native Streaming

- **Built on shards**
Each shard is a fixed unit of read/write capacity
- **Provisioned or on-demand mode**
Set shard count yourself, or let AWS auto-scale it
- **24-hour to 365-day retention**
Configurable how long data sticks around
- **Consumed via SDK, KCL, Lambda**
Or services like Kinesis Data Analytics

Amazon MSK (Managed Streaming for Kafka)



Managed Apache Kafka

- **Real, open-source Apache Kafka**
Existing Kafka producers & consumers just work
- **Provisioned or MSK Serverless**
Size brokers yourself, or let it auto-scale
- **Speaks the standard Kafka protocol**
No proprietary APIs to learn
- **Compatible with the Kafka ecosystem**
Kafka Connect, Kafka Streams, ksqlDB

Key Differences

	Kinesis Data Streams	Amazon MSK
Underlying Tech	AWS-native (proprietary)	Apache Kafka (open source)
API / Protocol	Kinesis API / SDK	Standard Kafka protocol
Scaling	Shards — manual or on-demand	Brokers/partitions or Serverless
Ecosystem	AWS-native (Lambda, Firehose)	Kafka Connect, Streams, ksqlDB
Ops Overhead	Low — fully abstracted	Higher — more to configure
Portability	AWS only	Portable across clouds

When to Use Kinesis Data Streams



Building an AWS-native app

You want the simplest path to streaming



Tight AWS integration

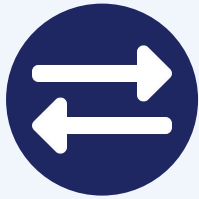
Lambda, Firehose & Kinesis Data Analytics out of the box



No existing Kafka expertise

Skip the learning curve entirely

When to Use Amazon MSK



Migrating existing Kafka

*Move to AWS without rewriting
producers/consumers*



Need Kafka ecosystem tools

*Kafka Connect for integrations, ksqlDB for
processing*



Portability matters

Avoid lock-in to a single cloud provider

Recap

- Kinesis Data Streams: AWS-native, simple, tightly integrated.
- Amazon MSK: managed Apache Kafka, powerful & portable.
- Choose based on your team's needs: simplicity vs. Kafka compatibility.

Thanks for watching!